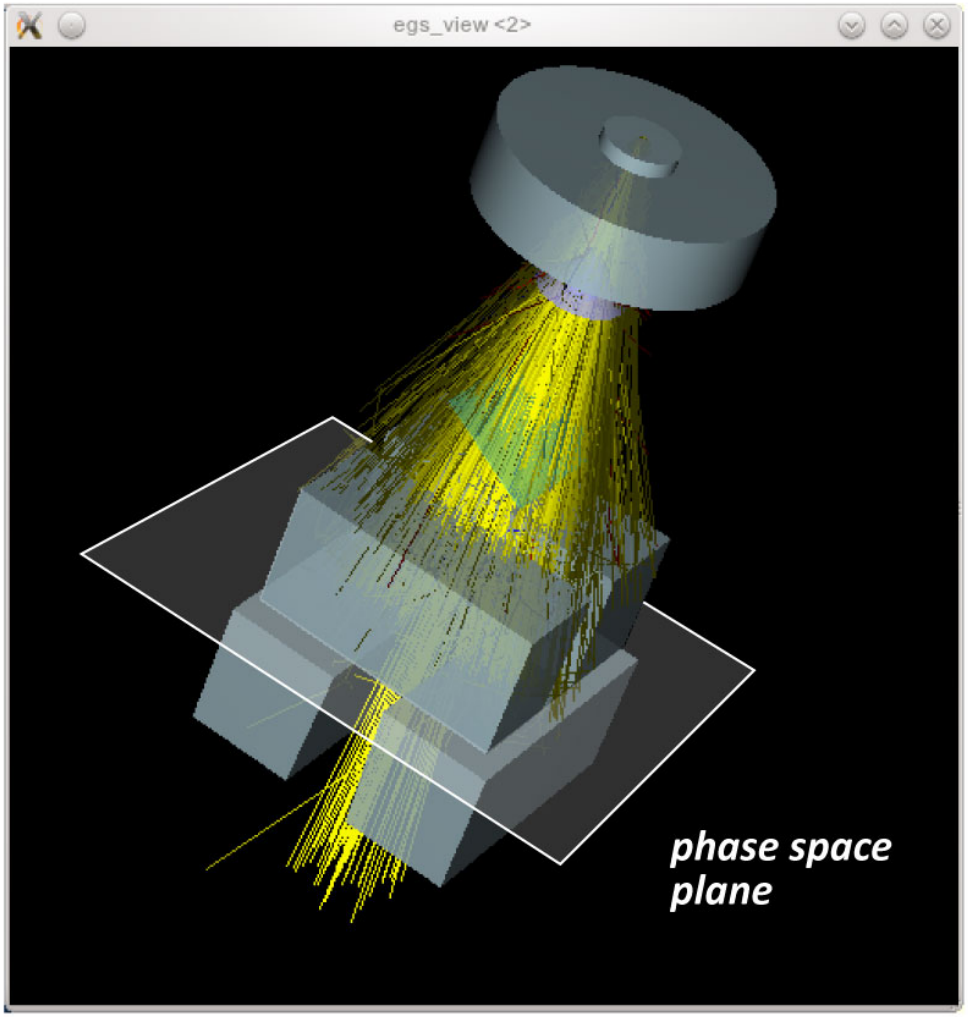
ICTP-SAIFR EGSnrc Monte Carlo School for Medical Physics
IFT-UNESP, São Paulo, Brazil
March 30 – April 3, 2026

LECTURE 10 · EGSnrc course

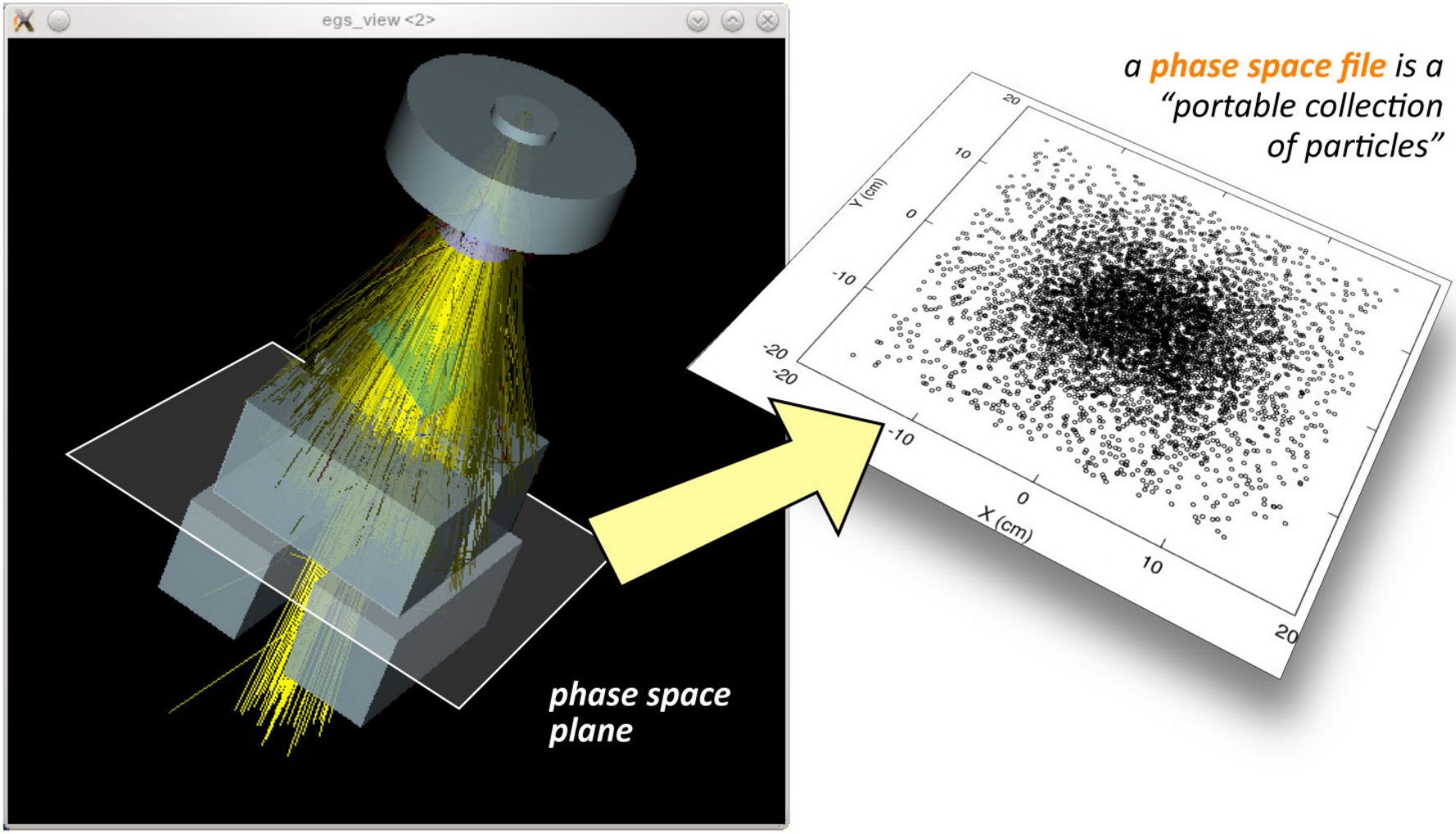
Phase space files

Frédéric Tessier · National Research Council Canada, Metrology Research Centre

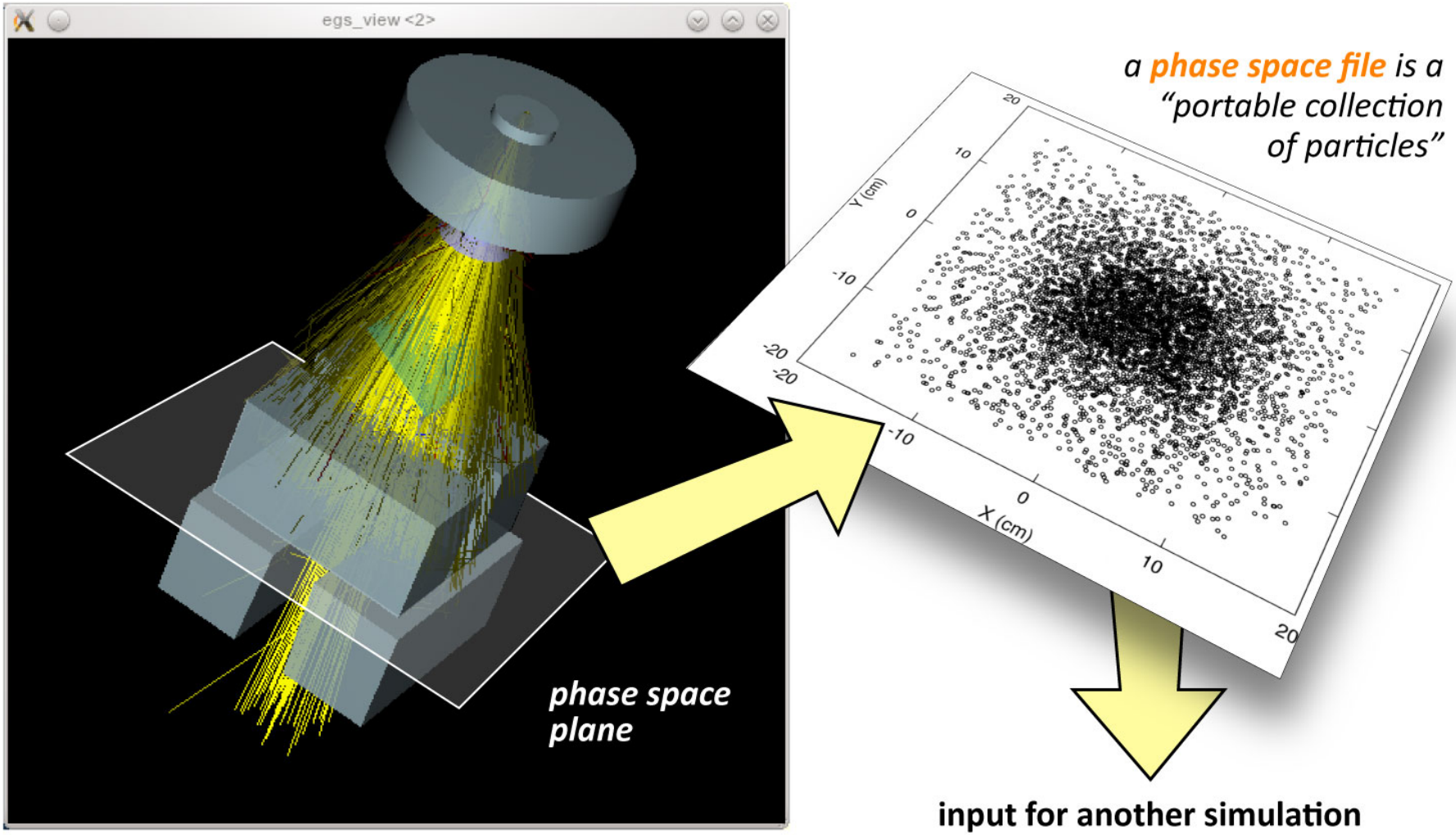
The BIG idea



The BIG idea


egs_view <2>
phase space plane
a phase space file is a "portable collection of particles"
Y (cm)
X (cm)
20
10
0
-10
-20

The BIG idea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phase space plane
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Y (cm)
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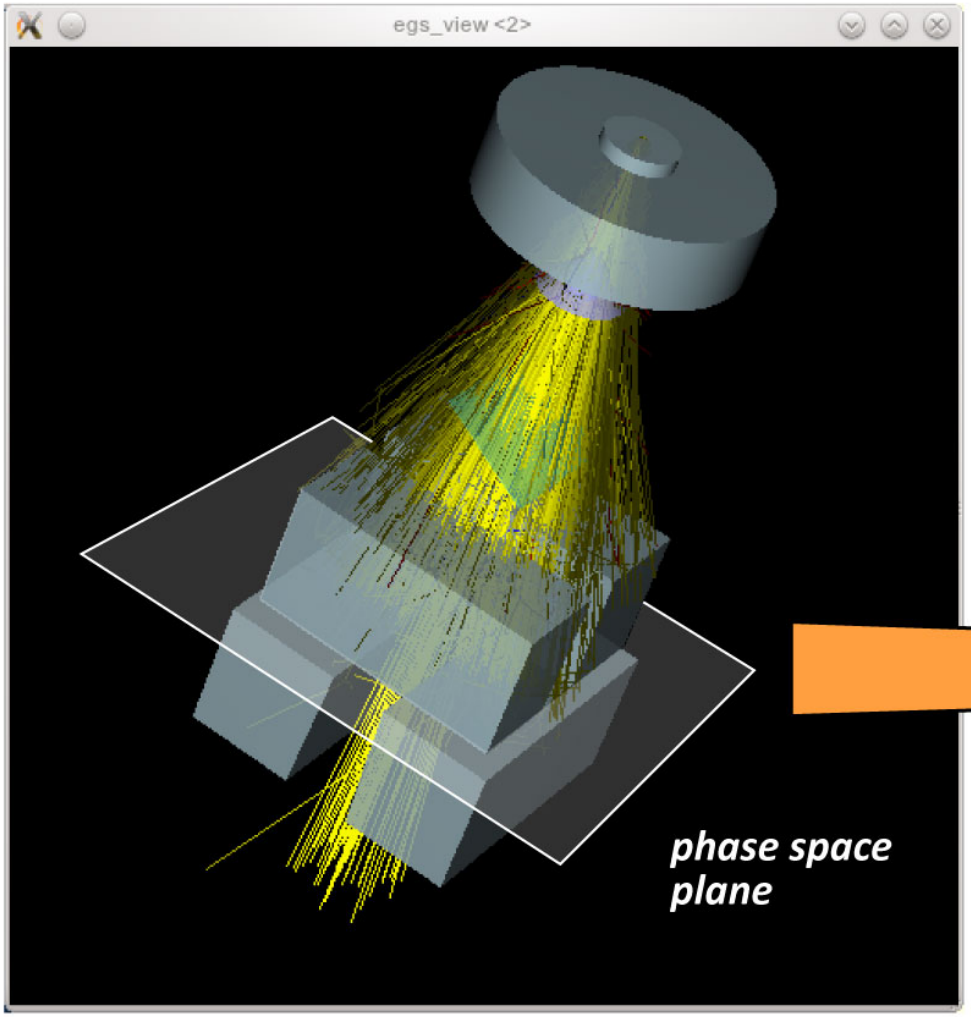
input for another simulation

The BIGGER idea



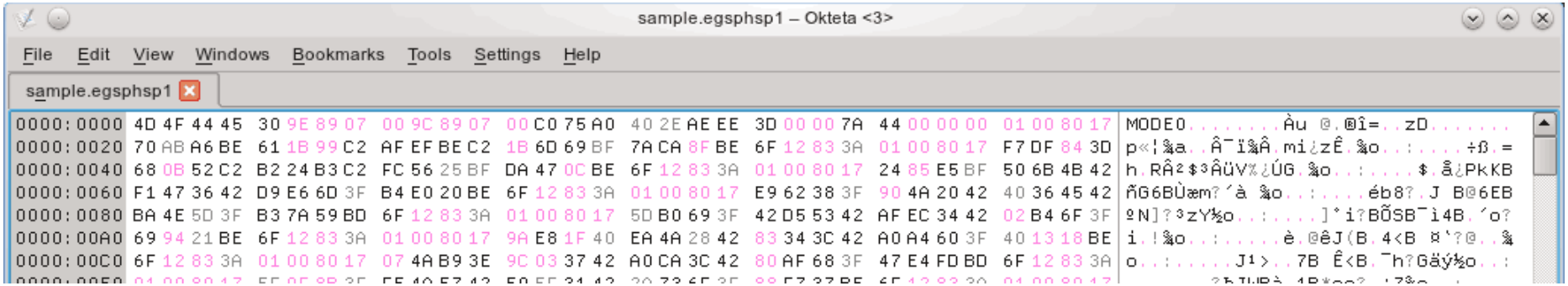
use BEAMnrc simulation ***directly***
as a source for another simulation

simulations call the BEAMnrc
shared library when they
need more particles

another simulation

Phase space files contain particle data

Binary format :-(



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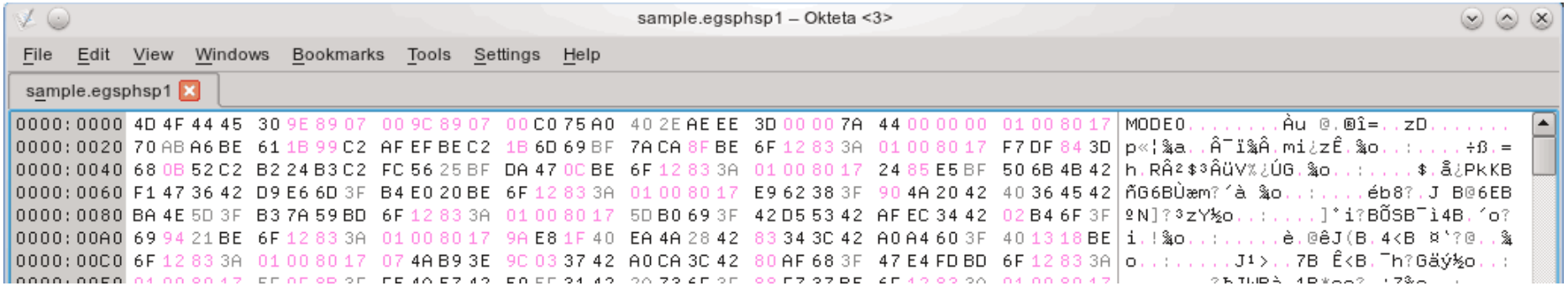
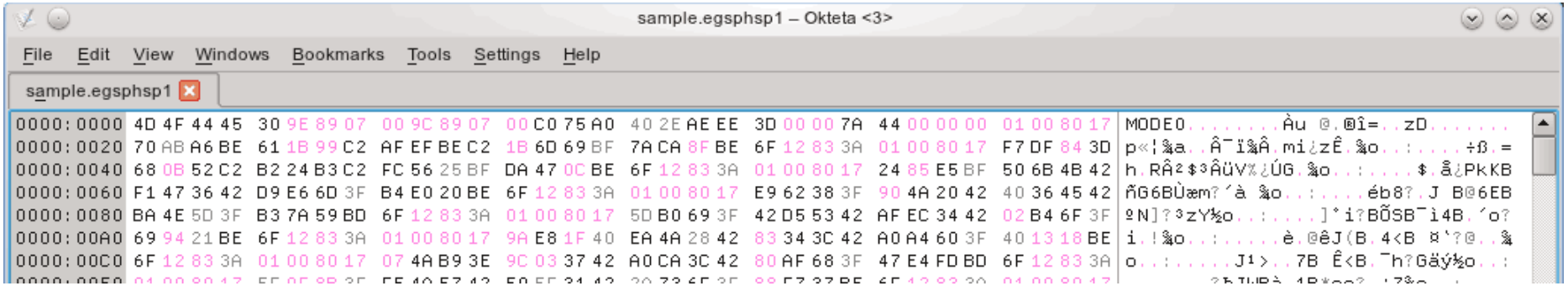


There is one phase space file for each scoring plane

`inputfile.egsphsp#`
where `#=1,2,3,...`

Phase space files contain particle data

Binary format :-(



There is one phase space file for each scoring plane

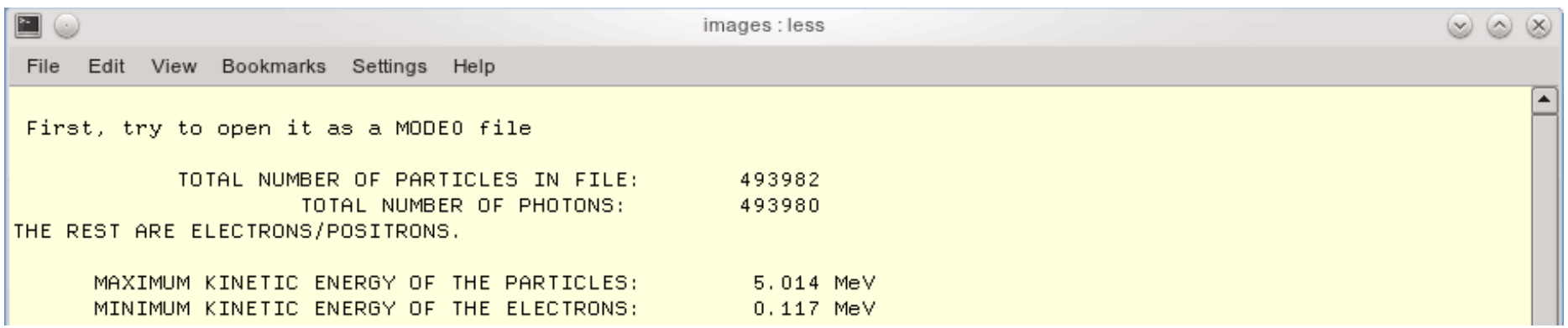
`inputfile.egsphsp#`
where `#=1,2,3,...`

Mind the file size!

Phase space files can rapidly grow to fill an entire hard drive. Test for a small number of histories in order to gauge the size of the phase space file for your final simulation.

A phase space file starts with a header

MODE_RW	MODE0 (no ZLAST) or MODE2 (with ZLAST)
NPPHSP	total number of particles in file
NPHOTPHSP	total number of photons in file
EKMAXPHSP	maximum kinetic energy of particles
EKMINPHSPE	minimum kinetic energy of electrons
NINCPHSP	number of particles incident from original source



A phase space file is a list of particles

For each particle

```
LATCHTMP [-]ESHORT X Y U V W WEIGHTTMP [ZLAST]
```

where:

`LATCHTMP`	LATCH value, charge (bits 29-30) and NPASS (bit 31)
`ESHORT`	total energy; set negative for new history
`X`	position along X axis (cm)
`Y`	position along Y axis (cm)
`U`	direction cosine with respect to X axis
`V`	direction cosine with respect to Y axis
`W`	direction cosine with respect to Z axis
`WEIGHTTMP`	statistical weight of particle, and sign of direction along the Z axis
`ZLAST`	**photons:** Z position of the last interaction; **electrons:** Z position where electron or its ancestor was set in motion by a photon

A phase space file is a list of particles

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```
 First, try to open it as a MODE0 file

            TOTAL NUMBER OF PARTICLES IN FILE:       493982
                     TOTAL NUMBER OF PHOTONS:        493980
THE REST ARE ELECTRONS/POSITRONS.

      MAXIMUM KINETIC ENERGY OF THE PARTICLES:        5.014 MeV
      MINIMUM KINETIC ENERGY OF THE ELECTRONS:        0.117 MeV
 # OF INCIDENT PARTICLES FROM ORIGINAL SOURCE:       1000.0

 ENERGY  IQ     X       Y      U      V     W    WEIGHT       LATCH (set=1, not set=0)

  0.326   0 -76.553 -95.468 -0.912 -0.281  0.300 1.000E-03  10111 10000000000000000000001
  0.065   0 -52.511 -89.572 -0.646 -0.137  0.751 1.000E-03  10111 10000000000000000000001
  1.793   0  50.855  45.570  0.929 -0.157  0.334 1.000E-03  10111 10000000000000000000001
  0.720   0  40.073  49.303  0.864 -0.053  0.500 1.000E-03  10111 10000000000000000000001
  0.913   0  52.958  45.231  0.936 -0.158  0.314 1.000E-03  10111 10000000000000000000001
  2.499   0  42.073  47.051  0.878 -0.149  0.456 1.000E-03  10111 10000000000000000000001
  0.362   0  45.754  47.198  0.909 -0.124  0.398 1.000E-03  10111 10000000000000000000001
  1.086   0  53.823  44.344  0.935 -0.180  0.305 1.000E-03  10111 10000000000000000000001
  3.718   0  46.349  47.150  0.913 -0.123  0.390 1.000E-03  10111 10000000000000000000001
  0.176   0  13.763  47.468 -0.668 -0.118  0.735 1.000E-03  10111 10000000000000000000001
  1.735   0  46.113  75.410  0.660  0.695  0.285 1.000E-03  10111 10000000000000000000001
  2.277   0  57.261  32.035  0.857 -0.450  0.252 1.000E-03  10111 10000000000000000000001
  1.012   0  41.756  46.688  0.872 -0.166  0.460 1.000E-03  10111 10000000000000000000001
  0.157   0  28.377  44.695  0.401 -0.450  0.798 1.000E-03  10111 10000000000000000000001
  0.636   0  43.376  46.380  0.886 -0.169  0.432 1.000E-03  10111 10000000000000000000001
  0.628   0  38.449  43.908  0.799 -0.321  0.508 1.000E-03  10111 10000000000000000000001
  0.401   0  23.484  45.936  0.011 -0.394  0.919 1.000E-03  10111 10000000000000000000001
  0.423   0  41.789  48.657  0.882 -0.079  0.465 1.000E-03  10111 10000000000000000000001
```

Changing where phase space files are written

1. Add the following input to your `.egsinp` file (either just before or just after the other EGSnrc inputs):

   ```
   :start user inputs:

   PHSP OUTPUT DIRECTORY= /full/path/to/new/output/directory

   :stop user inputs:
   ```

2. To change the destination for phase space files globally, edit the file `$OMEGA_HOME/beamnrc/beamnrc_user_macros` and change the macro

   ```
   REPLACE {$DIRECTORY-FOR-PHSP} WITH {$cstring(egs_home)//$cstring(user_code)};
   ```

   to read

   ```
   REPLACE {$DIRECTORY-FOR-PHSP} WITH {'/full/path/to/new/output/directory'};
   ```

   Don't forget to recompile your accelerator(s) for this change to take effect!

IAEA format for phase space files

International Atomic Energy Agency
Nuclear Data Services
Section Données Nucléaires, AIEA

International Advisory Committee (IAC)

R. Jeraj
I. Kawrakow
C.-M. Ma
D.W.O. Rogers
F. Sanchez-Doblado
J. Sempau
J. Seuntjens
J.V. Siebers
P. Andreo

Mailing Lists

Send mail to all members of the IAC

Register to the IAEA PHSP mailing list

Medical Portal

Atomic and nuclear data for medical applications

IAEA NAPC/NDS

Nuclear Data Section

IAEA NAHU/DMRP

Phase-space database for external beam radiotherapy

IAEA NAPC Nuclear Data Section
IAEA NAHU Dosimetry and Medical Radiation Physics Section

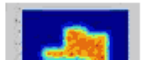
Project Officer: Roberto Capote

Objective: To build a database and disseminate representative phase-space data of accelerators and Co-60 units used in medical radiotherapy by compiling existing data that have been properly validated.

NEWS

WE ARE WA

How to produce and submit phase-space data: The IAEA phsp format was designed to cover both phase-space files and event generators (see phsp contents). We have implemented the IAEA phsp format in a set of read/write routines (Updated: September 2013, see readme file). Native IAEA phsp format is available in EGSnrc and PENELOPE Monte Carlo codes. Geant4 interface to use the native IAEA phsp format is also available. Once the validated phsp data is produced and documentation is published, you may submit your phsp for review using the upload link here.



How to download phase-space data: You have to select a phsp data type among Co-60 source, linac electron or linac photon phsps. For photon and electron PHSPs you may download the header first to decide which data you want to retrieve. Once decided you should download the PHSP data from the corresponding sub-directory. Please note that the first time access to the selected subdirectory could be slow.

Tech. Report

IAEA-NDS-0484

PHSP format

List of PHSP variables

PHSP Header

How to fill header ...

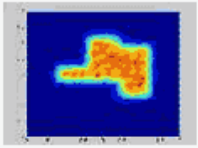
PHSP upload

Upload files

PHSP to review

Files to review

PHSP database



1. Co-60 phsps

IAEA format for phase space files

- Can make use of the International Atomic Energy Agency (IAEA) online accelerator phase space database: `www-nds.iaea.org/phsp/`.
- More details on this format are given by Capote et al. in IAEA report INDC (NDS)-0484.
- In order to be able to read and write in IAEA phase space format, EGSnrc and BEAMnrc must be configured on a system with a **working C++ compiler.**
- If the system is properly configured, then the IAEA phase space handling routines are automatically compiled for you in the shared object

  `$HEN_HOUSE/egs++/dso/iaea_phsp.so`

The IAEA format specifies a separate header file

The **text format** `.IAEAheader` file contains:

`$CHECKSUM`	size of phase space file.
`$RECORD_CONTENTS`	description of data stored
`$RECORD_CONSTANT`	values which are constant, e.g., Z of scoring plane for BEAMnrc
`$RECORD_LENGTH`	size of each particle data record, in bytes
`$BYTE_ORDER`	"1234" for little endian, "4321" for big endian
`$ORIG_HISTORIES`	number of primary histories
`$PARTICLES`	total number of particles
`$PHOTONS`	number of photons
`$STATISTICAL_INFORMATION_PARTICLES`	stats for each particle type
`$STATISTICAL_INFORMATION_GEOMETRY`	min/max in X, min/max in Y

The IAEA format specifies a separate header file

images : less

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```
$IAEA_INDEX:
1000   // Test header

$TITLE:
PHASESPACE in IAEA format

$FILE_TYPE:
0

$CHECKSUM:
16296786

$RECORD_CONTENTS:
    1     // X is stored ?
    1     // Y is stored ?
    0     // Z is stored ?
    1     // U is stored ?
    1     // V is stored ?
    1     // W is stored ?
    1     // Weight is stored ?
    0     // Extra floats stored ?
    2     // Extra longs stored ?
    1     // Incremental history number stored in the extralong array [ 0]
    2     // LATCH EGS variable stored in the extralong array [ 1]

$RECORD_CONSTANT:
    -0.0000     // Constant Z

$RECORD_LENGTH:
```

The IAEA phase space is written in a data file

For each particle, the **binary format** `.IAEAphsp` data file contains:

`type`	particle type, and sign of direction cosine with respect to Z
`E`	**KINETIC** energy of the particle
`X`	position along X axis (cm)
`Y`	position along Y axis (cm)
`U`	direction cosine with respect to X axis
`V`	direction cosine with respect to Y axis
`W`	direction cosine with respect to Z axis
`WT`	statistical weight of particle
`n_stat`	number of primary histories since previous particle scored
`LATCH`	value of `LATCH` variable
`ZLAST`	**photons:** Z position of the last interaction; **electrons:** Z position where electron was set in motion; This is only output if the input variable `IZLAST=1`

Reading and writing IAEA format phase space files

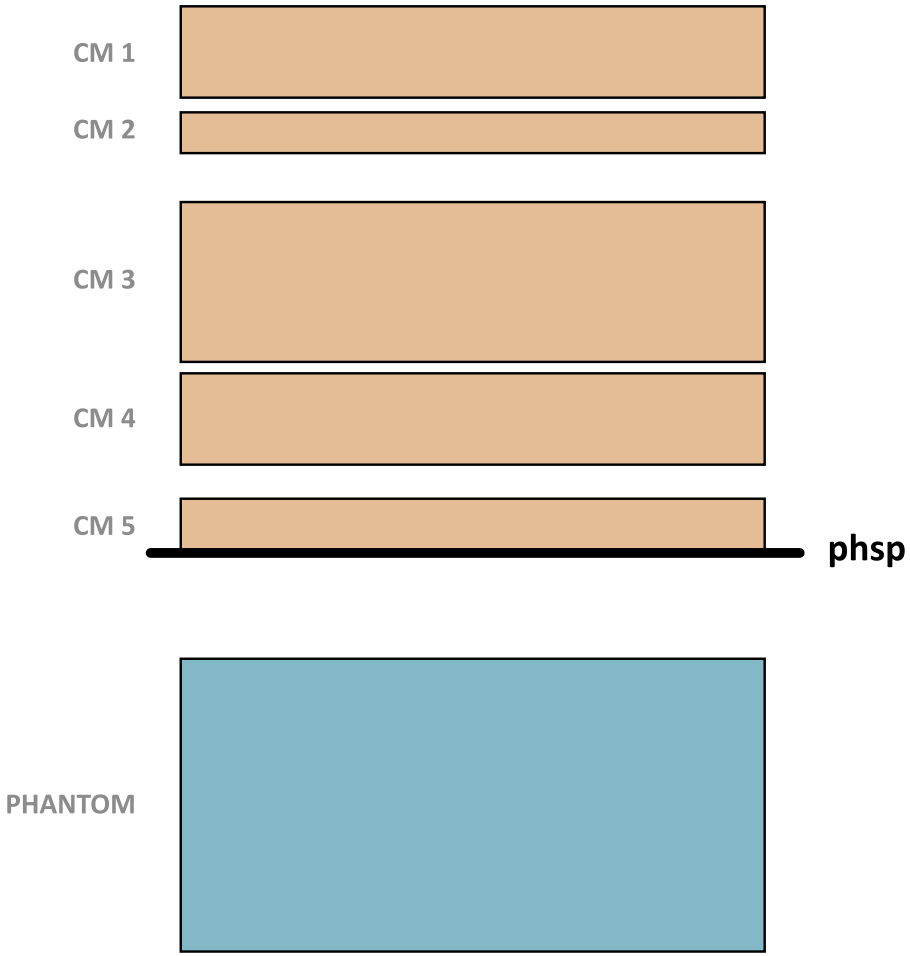
To write IAEA format phase space files from BEAMnrc:

- Set `IO_OPT=4`
- Remember that there are **two** files generated for each scoring plane. The naming scheme is `inputfile.#.IAEAheader` and `inputfile.#.IAEAphsp`, where `#` is the scoring plane number.

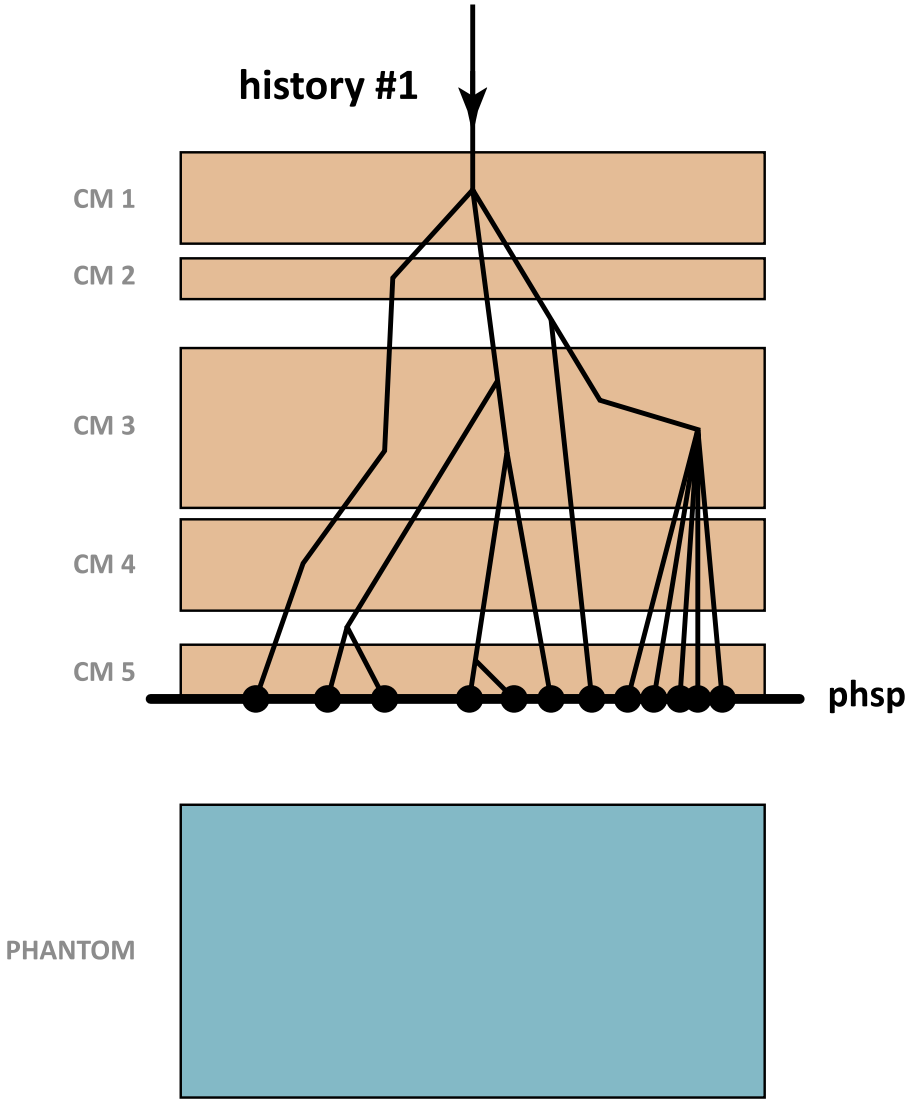
To read IAEA phase space files as sources in BEAMnrc, DOSXYZnrc and other EGSnrc applications:

- Input full name of the data file, including the `.IAEAphsp` extension, and the code takes care of the rest.
- Note that the code assumes that the separate `.IAEAheader` file is in the same directory as the data file.

Recycling and restarting

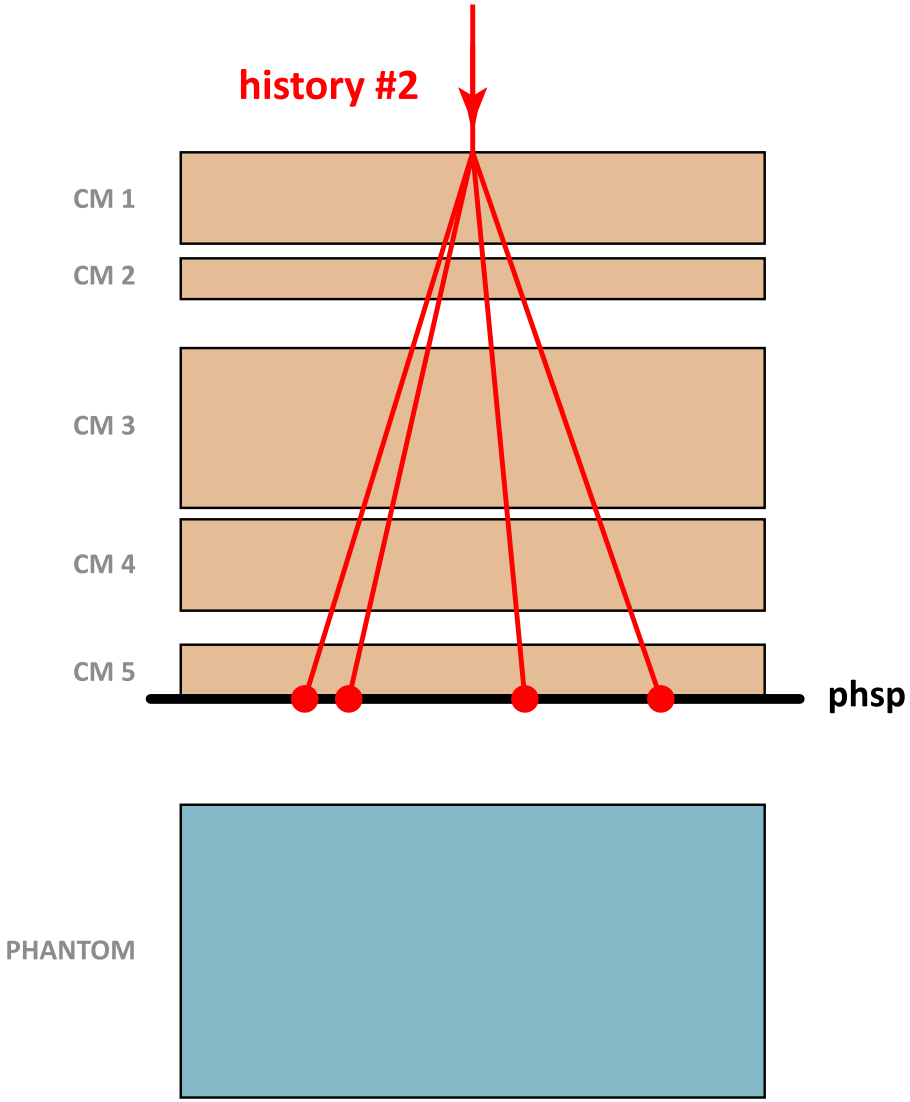


Recycling and restarting



ENERGY	IQ	X	Y	U	V	W	WEIGHT
-0.141	0	-22.109	1.214	-0.495	-0.083	0.865	1.000E-01
0.521	0	-15.575	19.570	-0.315	0.409	0.857	1.000E-01
0.224	0	-14.495	20.105	-0.287	0.424	0.859	1.000E-01
1.347	0	-12.743	27.185	-0.222	0.567	0.793	1.000E-01
1.981	0	-17.612	17.709	-0.370	0.359	0.857	1.000E-01
0.175	0	-10.898	27.875	-0.176	0.584	0.793	1.000E-01
1.758	0	-13.936	23.353	-0.263	0.493	0.829	1.000E-01
0.335	0	-9.783	-9.864	-0.182	-0.409	0.894	1.000E-01
1.736	0	16.136	20.923	0.483	0.404	0.777	1.000E-01
0.083	0	11.381	-6.120	0.403	-0.298	0.865	1.000E-01
0.231	0	33.138	25.116	0.695	0.392	0.603	1.000E-01
2.991	0	14.953	10.531	0.498	0.166	0.851	1.000E-01

Recycling and restarting



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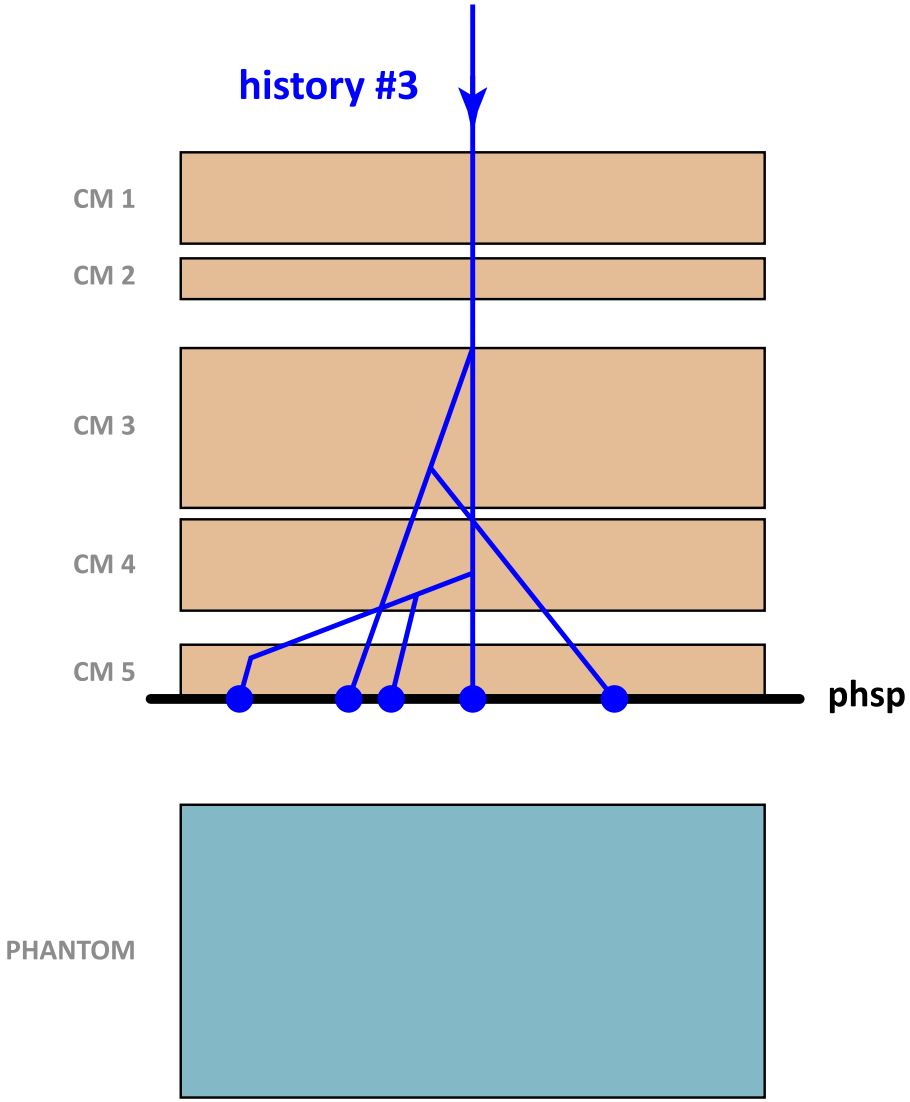
-2.273	0	8.072	21.841	0.205	0.538	0.818	1.000E-01
0.259	0	11.563	17.803	0.302	0.447	0.842	1.000E-01
0.385	0	-4.573	-15.985	-0.142	-0.479	0.866	1.000E-01
1.172	0	14.449	18.650	0.365	0.453	0.813	1.000E-01

Recycling and restarting



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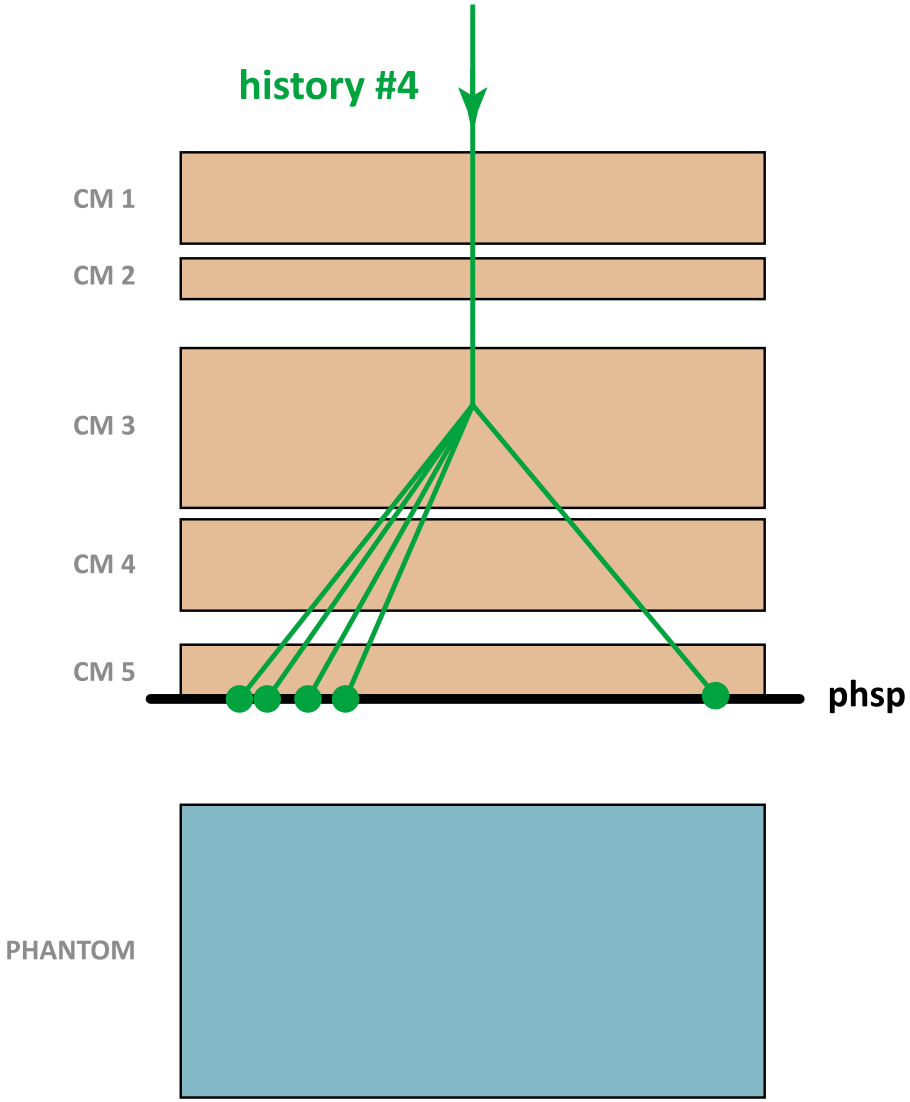
1.172	0	14.449	18.650	0.365	0.453	0.813	1.000E-01
-0.506	0	3.994	5.520	0.241	0.010	0.971	1.000E-01
0.224	0	-0.949	4.641	0.093	-0.017	0.995	1.000E-01
0.492	0	7.089	5.447	0.326	0.007	0.945	1.000E-01
1.258	0	-11.355	-5.180	-0.215	-0.301	0.929	1.000E-01
1.115	0	1.283	9.183	0.160	0.122	0.980	1.000E-01

Recycling and restarting



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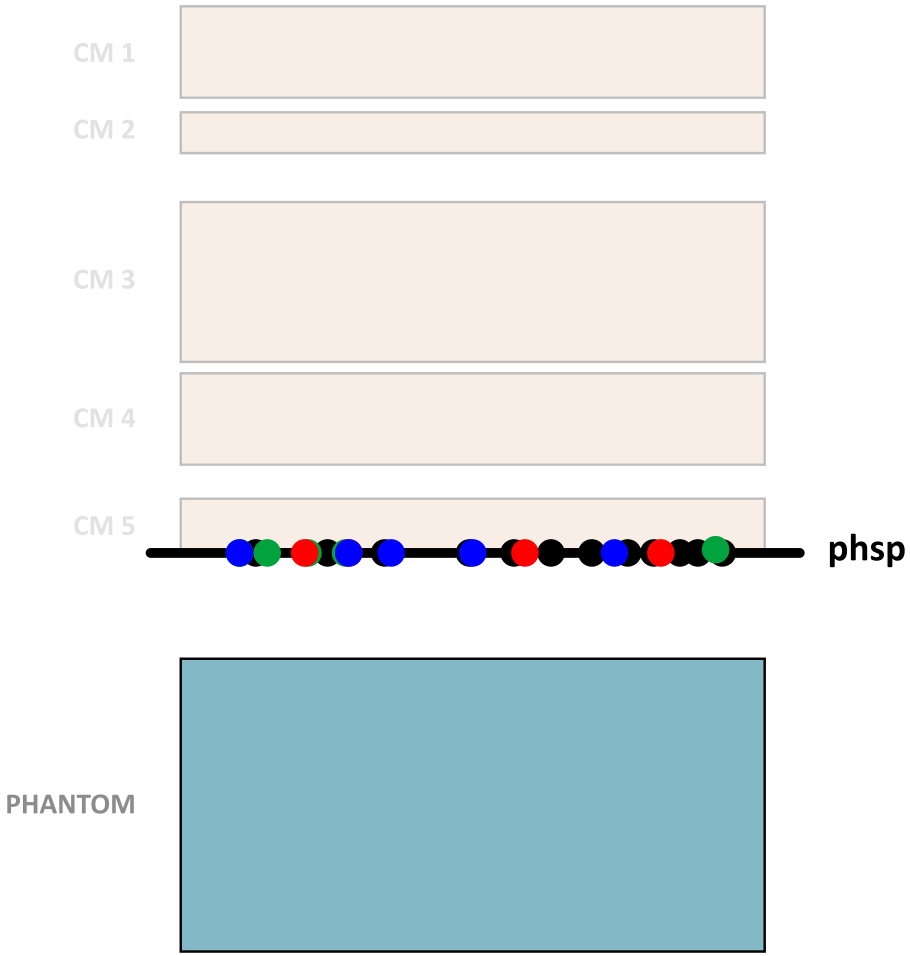
1.172	0	14.449	18.650	0.365	0.453	0.813	1.000E-01
-0.506	0	3.994	5.520	0.241	0.010	0.971	1.000E-01
0.224	0	-0.949	4.641	0.093	-0.017	0.995	1.000E-01
0.492	0	7.089	5.447	0.326	0.007	0.945	1.000E-01
1.258	0	-11.355	-5.180	-0.215	-0.301	0.929	1.000E-01
1.115	0	1.283	9.183	0.160	0.122	0.980	1.000E-01
-0.056	0	-13.623	11.405	-0.368	0.255	0.894	1.000E-01
0.203	0	-7.327	10.600	-0.315	0.264	0.912	1.000E-01
0.556	0	91.982	13.935	0.939	0.117	0.325	1.000E-01
0.363	0	19.035	-23.710	0.429	-0.571	0.700	1.000E-01
0.220	0	8.198	-8.152	0.252	-0.303	0.919	1.000E-01
(...)							

Recycling and restarting



ENERGY	IQ	X	Y	U	V	W	WEIGHT
-0.141	0	-22.109	1.214	-0.495	-0.083	0.865	1.000E-01
0.521	0	-15.575	19.570	-0.315	0.409	0.857	1.000E-01
0.224	0	-14.495	20.105	-0.287	0.424	0.859	1.000E-01
1.347	0	-12.743	27.185	-0.222	0.567	0.793	1.000E-01
1.981	0	-17.612	17.709	-0.370	0.359	0.857	1.000E-01
0.175	0	-10.898	27.875	-0.176	0.584	0.793	1.000E-01
1.758	0	-13.936	23.353	-0.263	0.493	0.829	1.000E-01
0.335	0	-9.783	-9.864	-0.182	-0.409	0.894	1.000E-01
1.736	0	16.136	20.923	0.483	0.404	0.777	1.000E-01
0.083	0	11.381	-6.120	0.403	-0.298	0.865	1.000E-01
0.231	0	33.138	25.116	0.695	0.392	0.603	1.000E-01
2.991	0	14.953	10.531	0.498	0.166	0.851	1.000E-01
-2.273	0	8.072	21.841	0.205	0.538	0.818	1.000E-01
0.259	0	11.563	17.803	0.302	0.447	0.842	1.000E-01
0.385	0	-4.573	-15.985	-0.142	-0.479	0.866	1.000E-01
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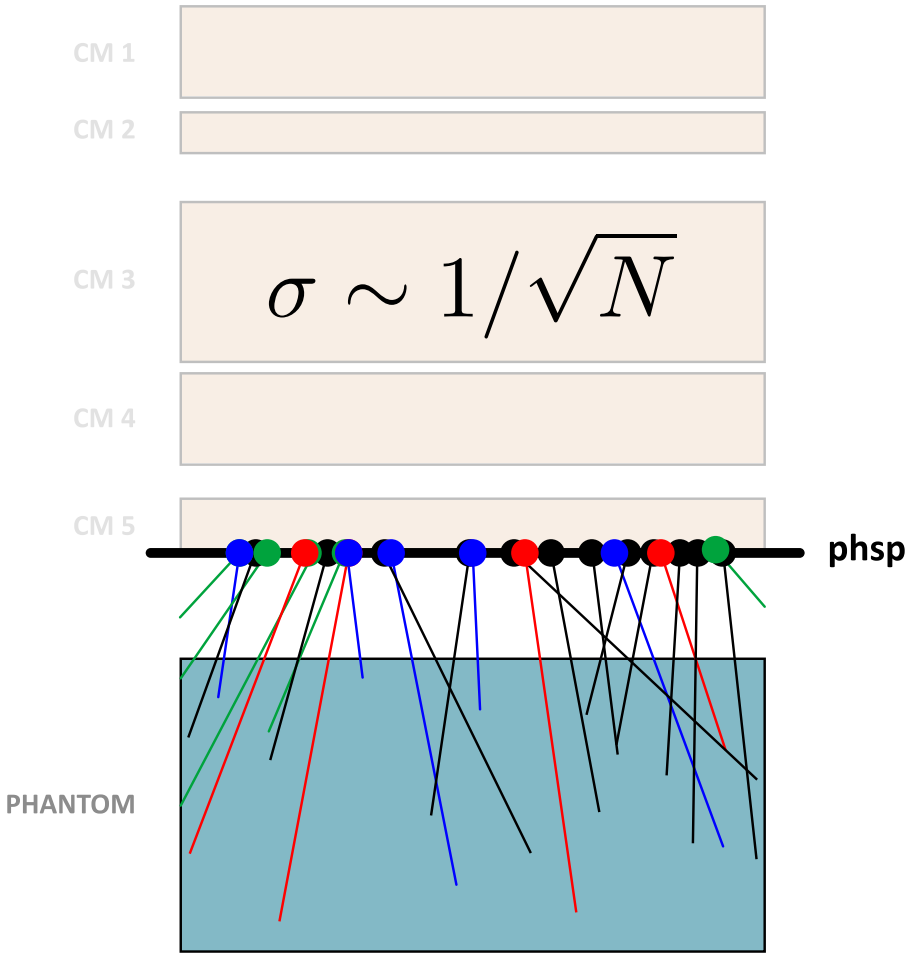
0.220	0	8.198	-8.152	0.252	-0.303	0.919	1.000E-01
(...)							

Recycling and restarting



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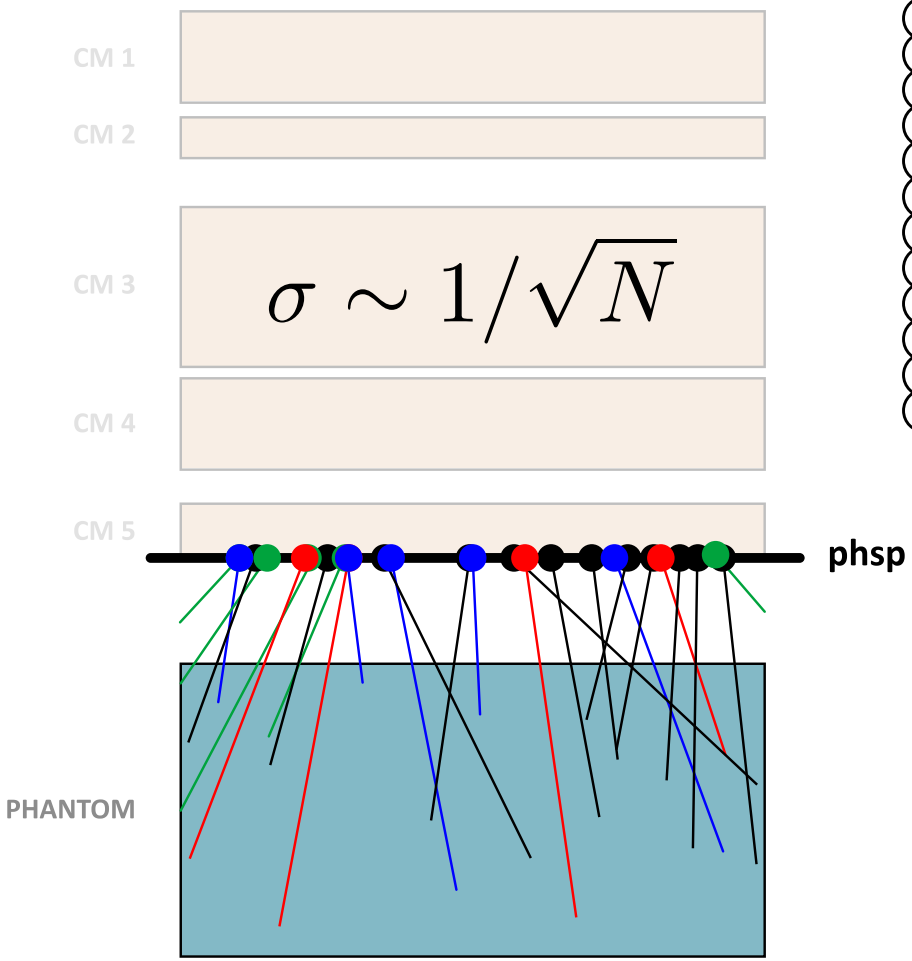
0.220	0	8.198	-8.152	0.252	-0.303	0.919	1.000E-01
(...)							

Recycling and restarting



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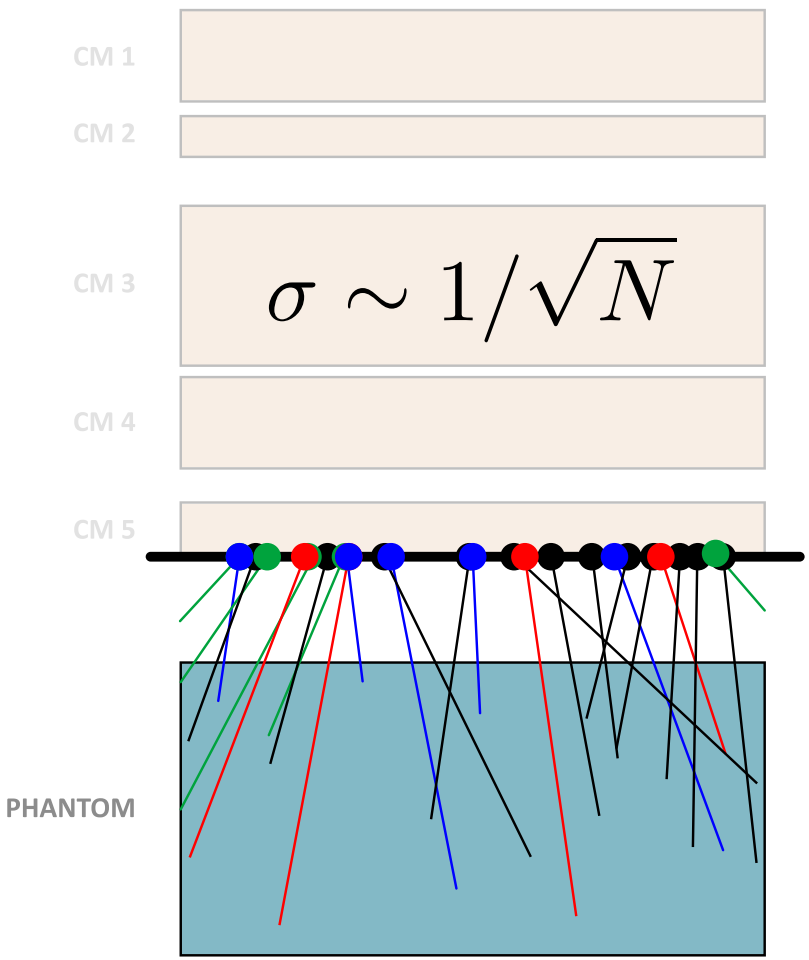
(...)							

Recycling and restarting



$$\sigma \sim 1/\sqrt{N}$$

×2

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0.224	0	-14.495	20.105	-0.287	0.424	0.859	1.000E-01
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0.231	0	33.138	25.116	0.695	0.392	0.603	1.000E-01
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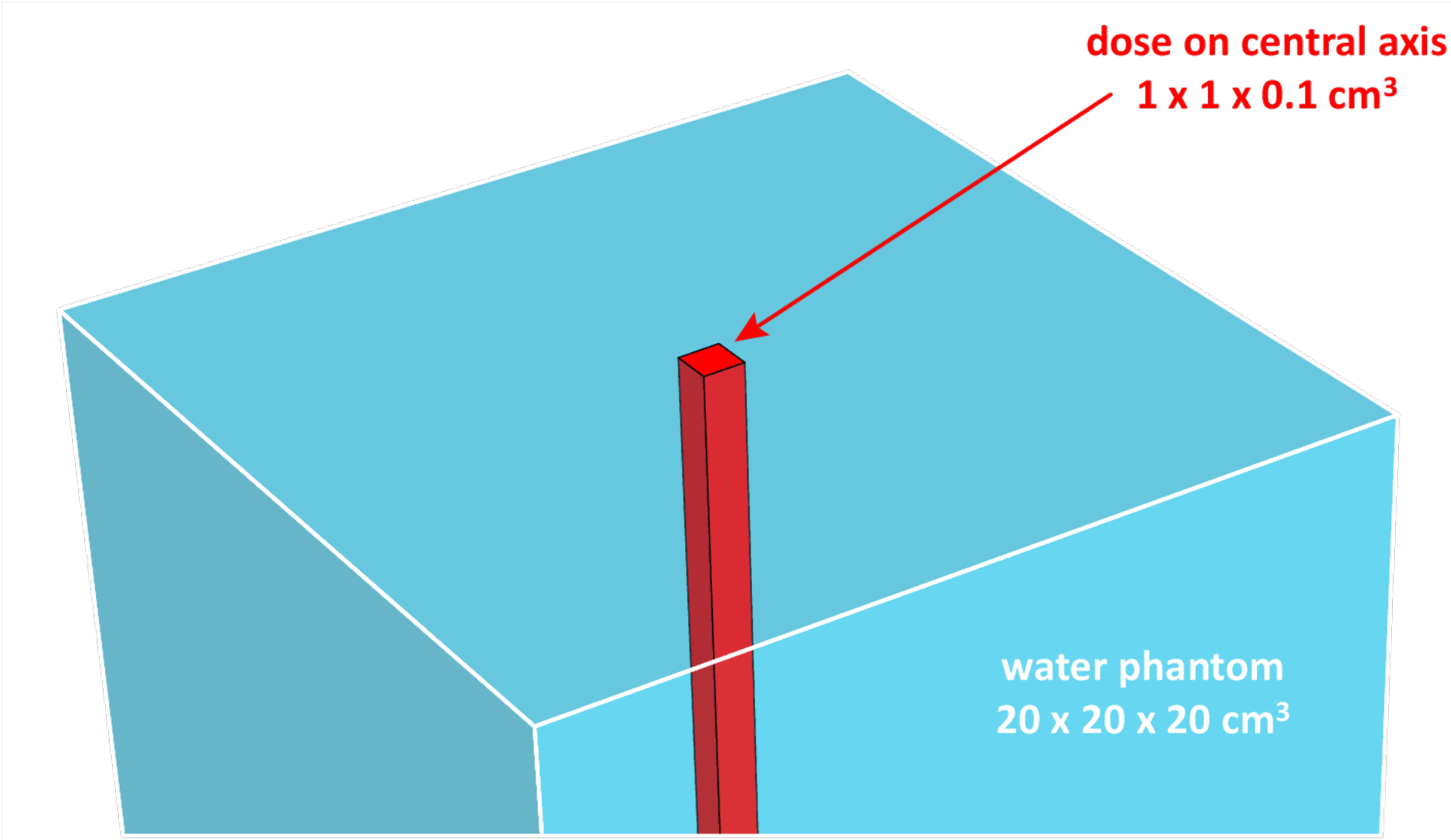
0.220	0	8.198	-8.152	0.252	-0.303	0.919	1.000E-01
(...)							

Executive summary (if you *must* reuse particles)

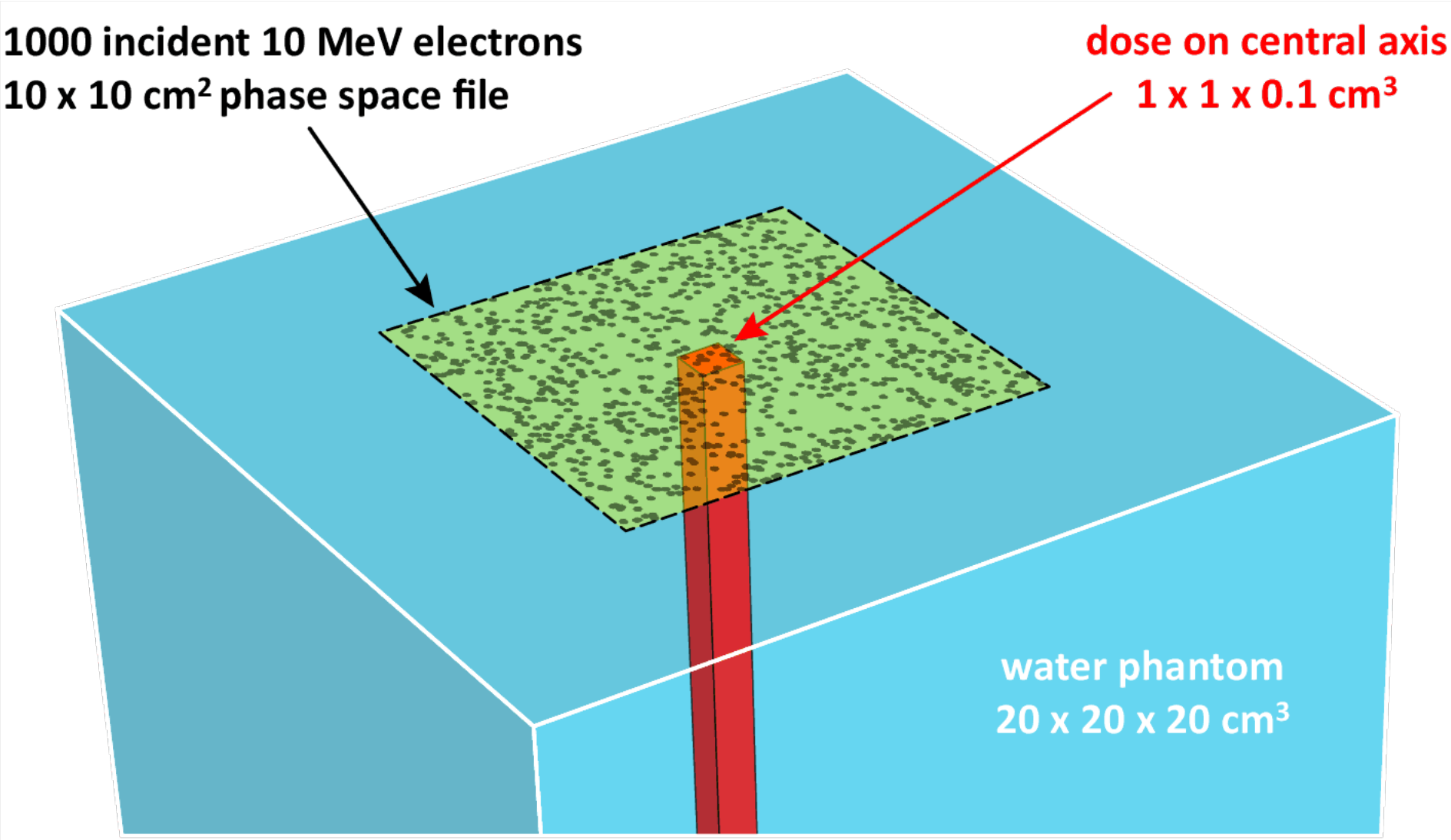
Recycling is good

Restarting is bad

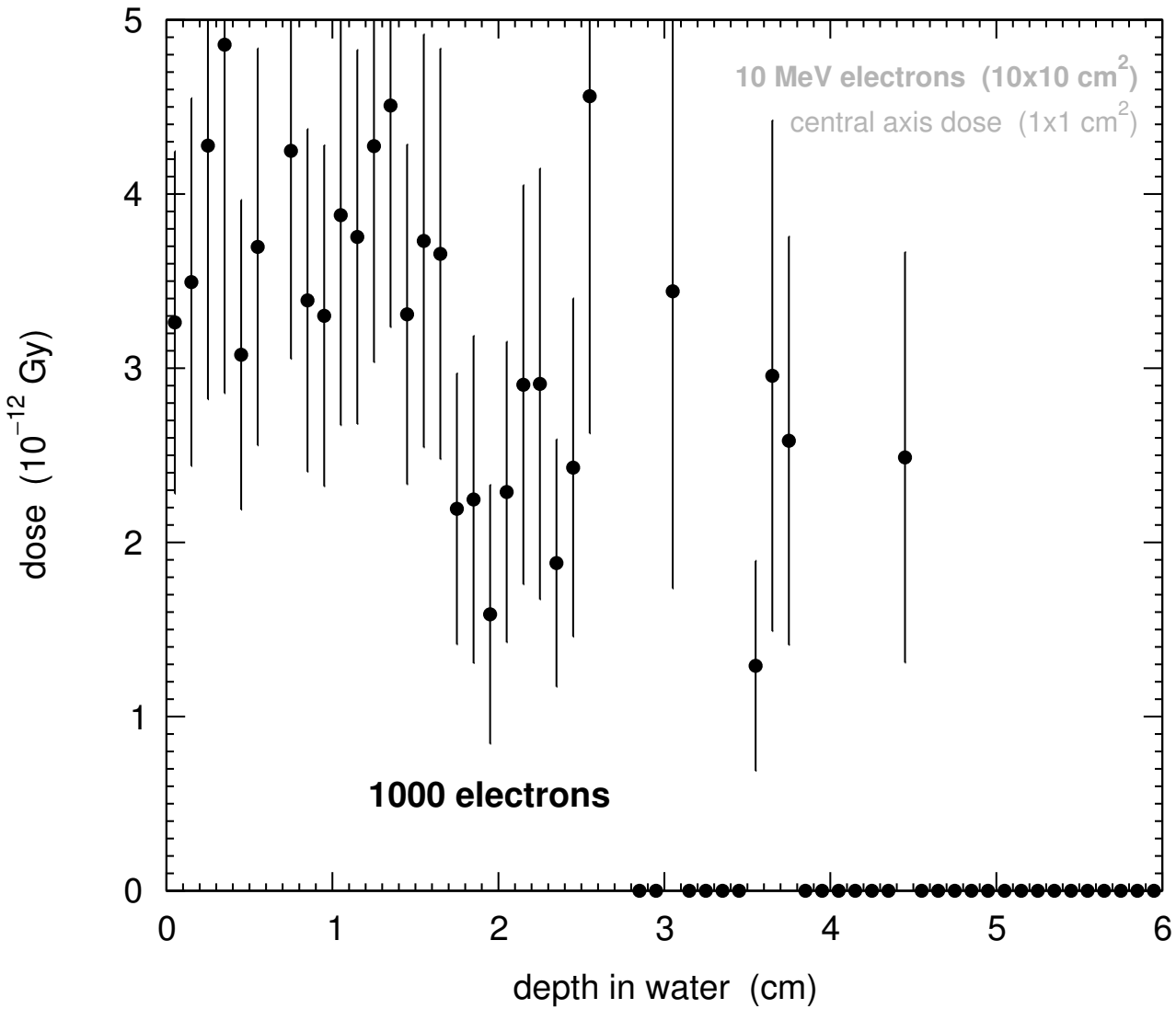
Example: electron depth-dose curve in water



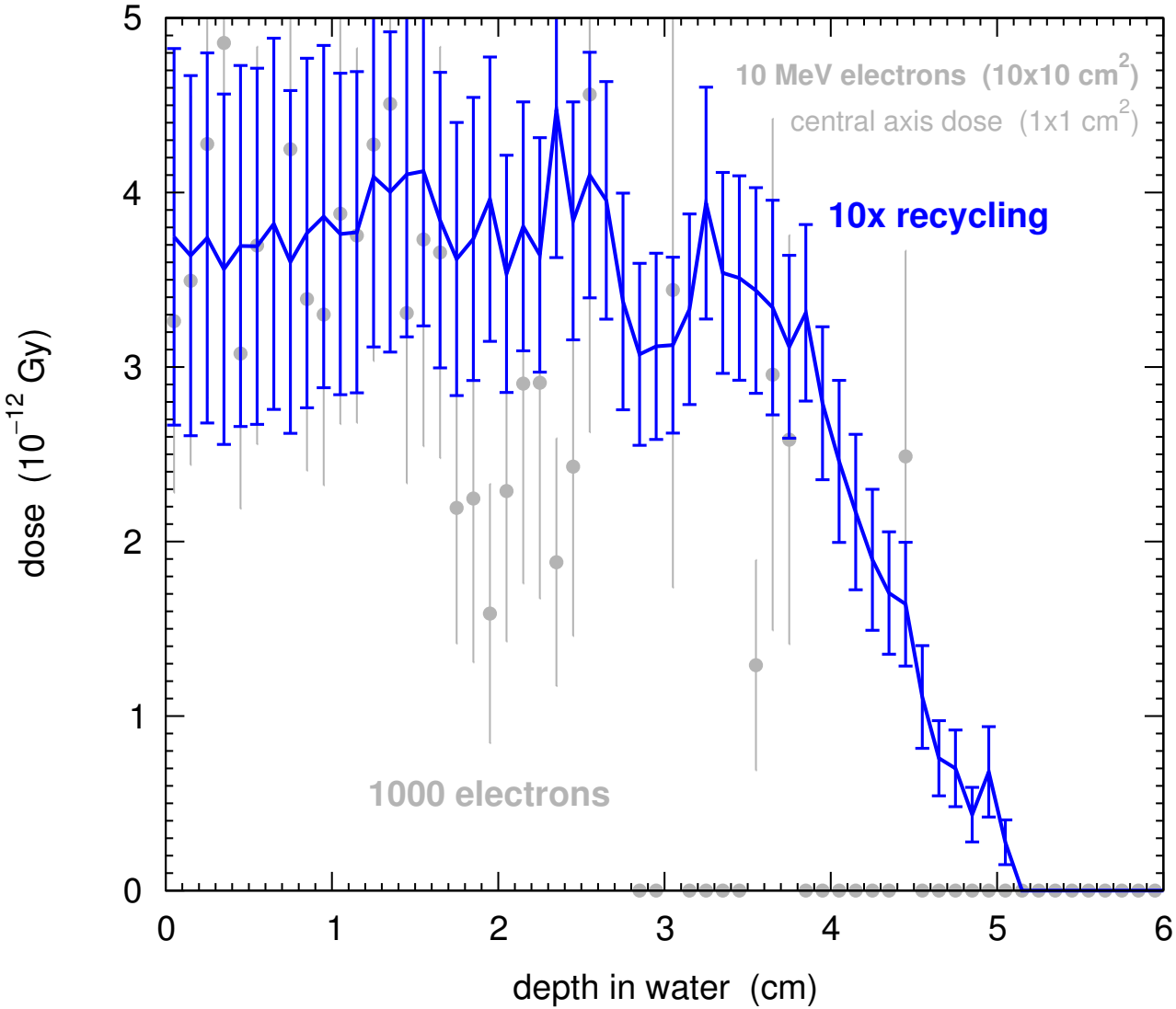
Example: electron depth-dose curve in water



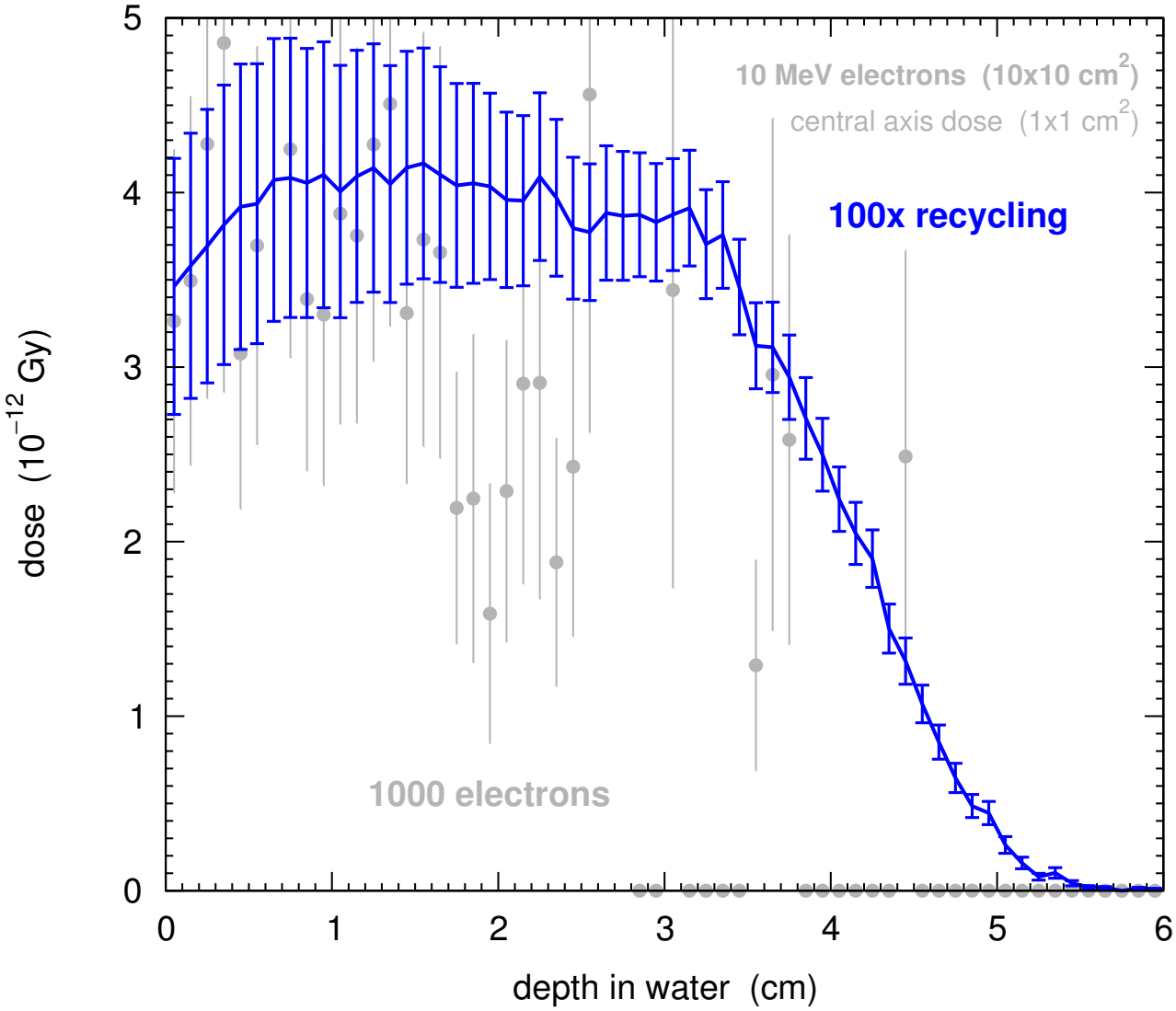
Example: electron depth-dose curve in water



Example: electron depth-dose curve in water



Example: electron depth-dose curve in water



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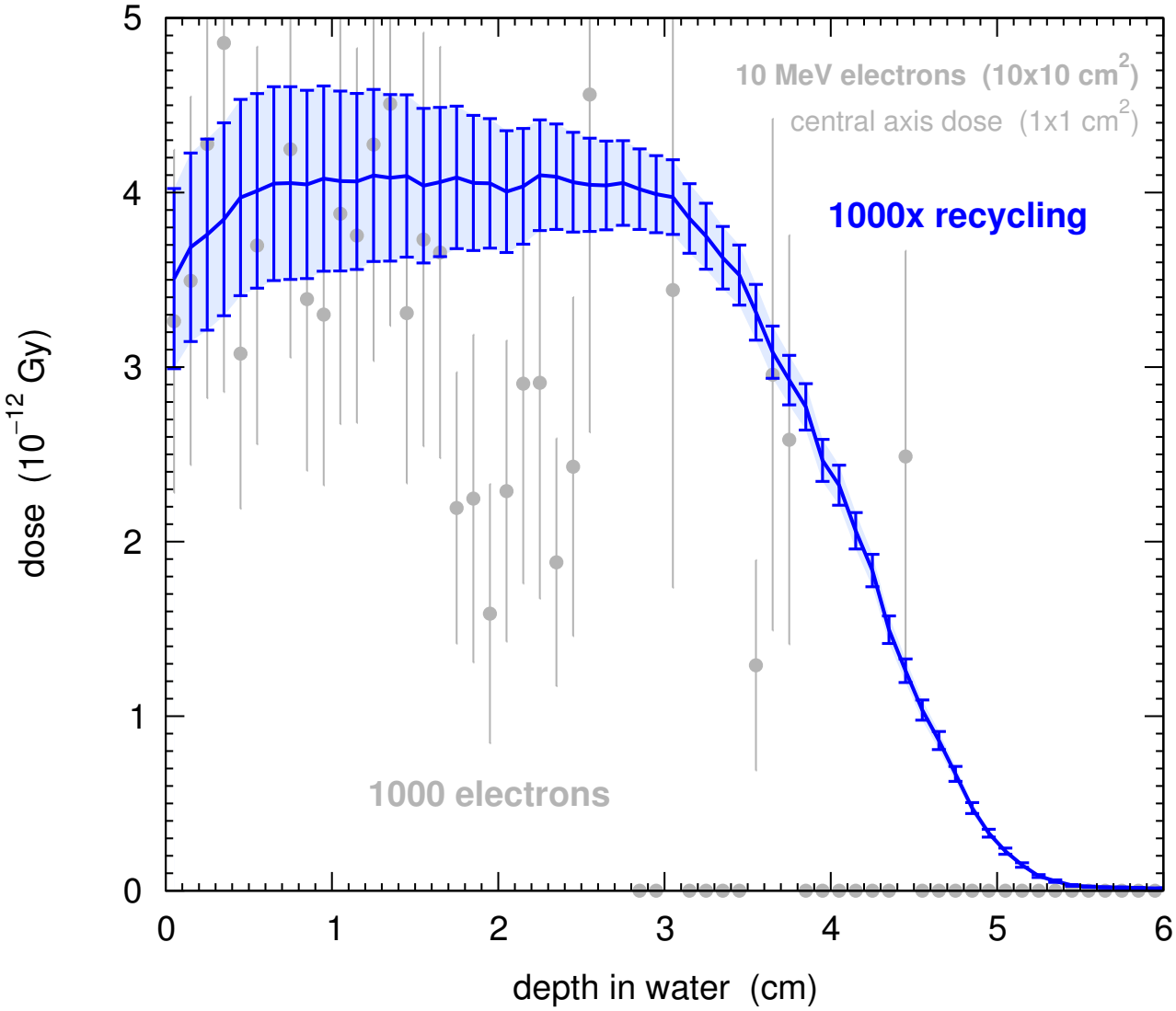
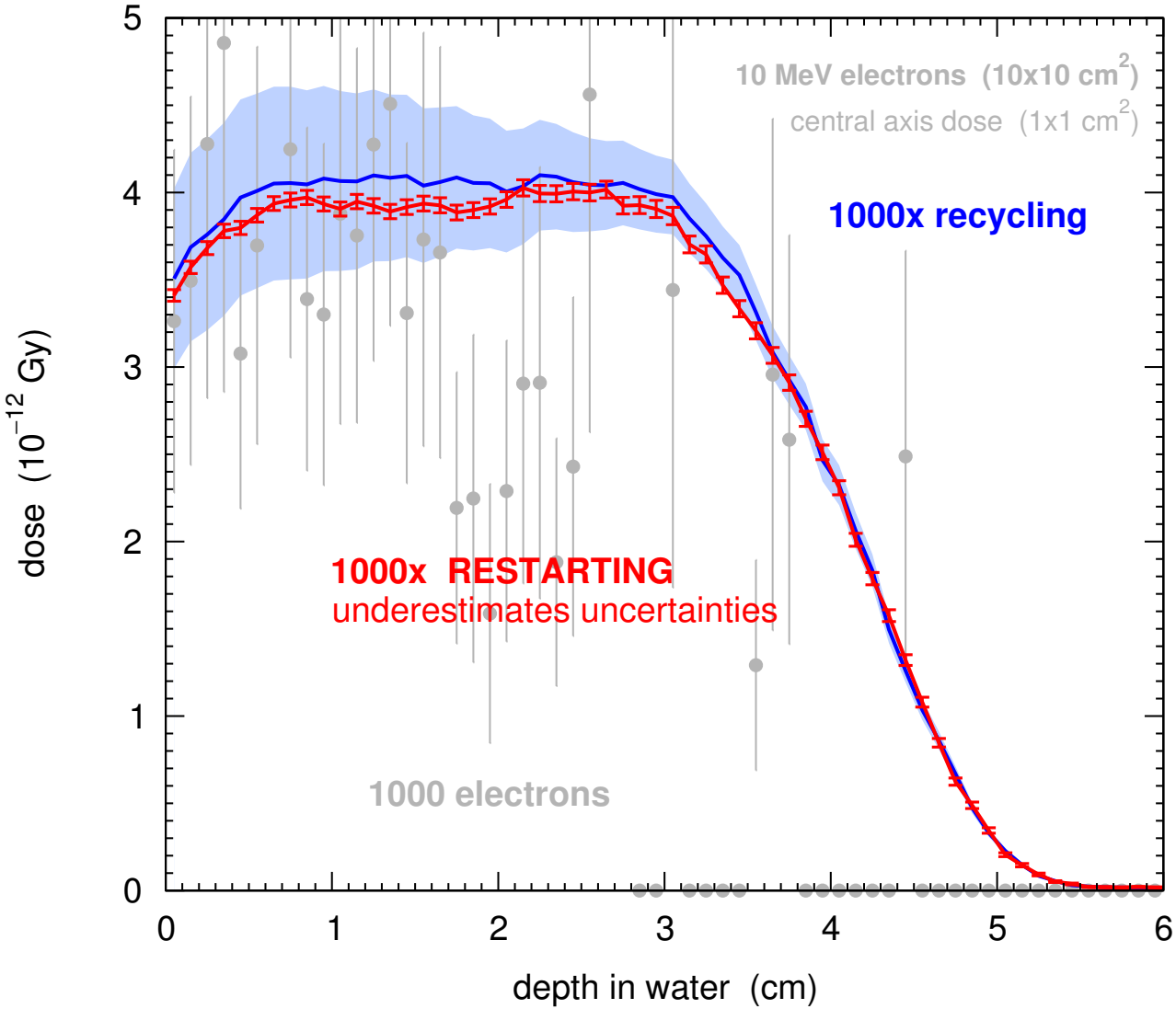
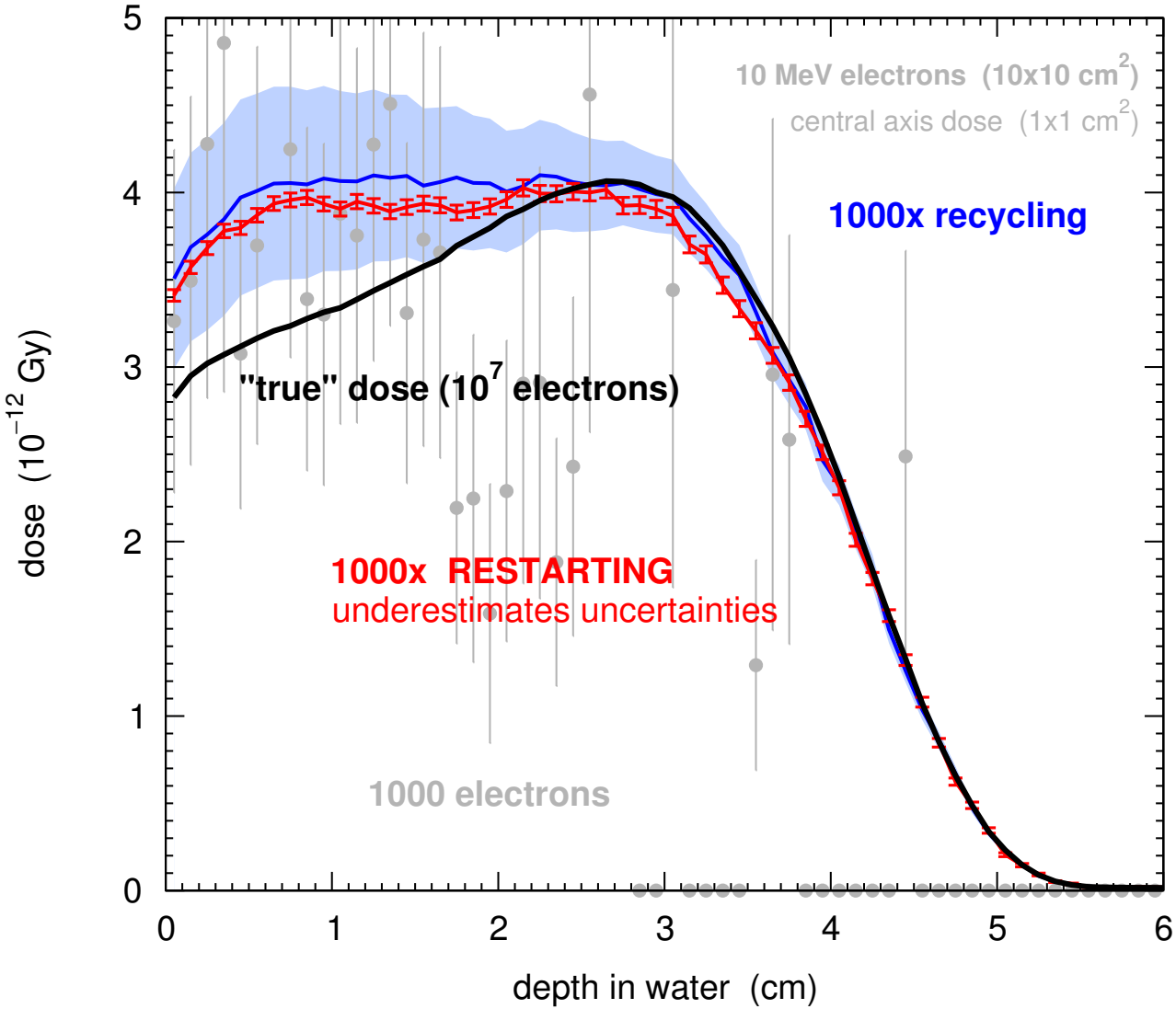


Example: electron depth-dose curve in water



Example: electron depth-dose curve in water



Conclusion

Restarting: stop.

Conclusion

Restarting: stop.

Recycling: slow down...

Conclusion

Restarting: stop.

Recycling: slow down...

More particles: go!